Team Name: 1684R

ICT2113

Project: The Abyss Navigator Suite

Name	Student ID
Muhammad Ridwan Putra Jasni	2401684
Chng Zong Han	2401892
Lin Yuhao	2400703
Muhammad Mikhail Bin Mazlan	2402298
Kannan s/o Rajamohan	2401517

1. Project Description

1.1 Project Name

Abyss Navigator Suite – Unmanned Deep-Sea Submersible Fleet Control System

1.2 Customer Organization

Triton Deepwater is a safety-first marine research and deep-sea oil & gas company. We define the operational intent, constraints, and priorities that any proposed solution must satisfy in order to be considered for tender.

1.3 Vendor Organization

The selected vendor will be responsible for proposing and delivering a software-based control system that satisfies the requirements stated in this specification. No vendor has been pre-selected; this specification is issued prior to tender release and is intended to guide all prospective bidders equally.

1.4 Context

Triton Deepwater conducts operations in extreme deep-sea environments where human presence introduces unacceptable risk. In alignment with our safety-first operating principles, all future deep-sea missions are defined as unmanned and remotely supervised. The Abyss Navigator Suite is intended to support this operating model by enabling safe, autonomous, and accountable execution of deep-sea missions without onboard crew presence.

1.5 Project Goal

Our goal is to procure a safety-critical operational capability that enables autonomous deep-sea operations, supports controlled human intervention from the mothership when required, ensures reliable emergency recovery, and provides comprehensive operational traceability. This goal is defined in outcome terms rather than prescribed technical solutions, allowing vendors to propose appropriate designs during the tender process.

Achievement of these goals shall be assessed through mission outcomes, operational safety performance, emergency handling effectiveness, and post-mission traceability during acceptance testing and sea trials, within the project schedule defined in this specification.

2. Scope

We define the scope of this Project Specification to establish clear boundaries of responsibility and expectation for the forthcoming tender. The scope clarifies the operational domains and responsibilities that proposed solutions are expected to address, as well as areas that are explicitly excluded to prevent ambiguity during vendor evaluation.

2.1 In Scope

The scope of this project includes software-enabled capabilities that support unmanned deep-sea mission operations, including:

- Fleet-level oversight and coordination of unmanned deep-sea submersibles from a surface or mothership environment
- Support for autonomous mission execution with provision for supervised human intervention when required
- Integration, at the software level, with mission sensors supplied by us
- Monitoring and reporting of vehicle operational status and integrity
- Support for mission safety, emergency handling, and recovery coordination
- Recording of mission activity and operational decisions to support accountability and post-mission analysis

2.2 Out of Scope

The following items are explicitly excluded from the scope of this Project Specification:

- Human occupancy, life-support, or crew-related systems
- Monitoring of crew or personnel health
- Manufacture or supply of physical submersibles, sensors, hulls, propulsion systems, or other hardware
- Operation or provision of onboard manual piloting stations intended for human occupancy

3. Stakeholders

We identify the following stakeholder groups as having a direct or indirect interest in the Abyss Navigator Suite. These stakeholders represent operational, safety, engineering, and regulatory perspectives that have informed the requirements defined in this Project Specification.

- Triton Deepwater Fleet Operations Team
- Triton Deepwater Safety & Compliance Division
- Triton Deepwater Offshore Engineering Group
- Mothership-based pilots and mission controllers
- Maritime and international regulatory authorities (indirect)

4. Assumptions, Constraints, Regulations, and Operating Context

We operate in extreme deep-sea environments where human presence introduces unacceptable risk. The following assumptions, constraints, regulations, and operating contexts apply to all requirements in this Project Specification and are independent of any specific vendor solution.

4.1 Assumptions

- Missions are supervised by trained and authorised personnel operating from a surface or mothership environment.
- Primary mission sensors are supplied by us and made available for integration.

4.2 Business Constraints and Rules

- Deep-sea operations are conducted without human presence at extreme depths.
- Human exposure to extreme deep-sea environments is eliminated as a business policy.
- Operations take place under conditions characterised by extreme pressure, low visibility, and constrained communication.

4.3 Regulatory Obligations

- Operations comply with applicable maritime safety regulations.
- Operations comply with relevant data-governance and operational accountability requirements imposed by governing authorities.

These assumptions, constraints, and regulations define the operating reality within which all subsequent requirements must be satisfied.

5. Requirements Tree

Requirements in this Project Specification are organised using a **Contextually Relevant and Meaningful (CRaM)** requirements tree. The tree structures requirements by **granularity and intent**, ensuring consistent application across the total set of requirement clauses.

5.1 Level Definitions

- **Level 0** — Business context, constraints, regulations, and assumptions that may stand alone and constrain all lower-level requirements.
- **Level 1** — Business or IT requirements derived from exactly one Level 0 item.
- **Level 2** — Elaboration of Level 1 requirements where additional clarification is required.

5.2 Business Operations Requirements Tree

The Business Operations requirements define the operational outcomes, business rules, constraints, assumptions, and regulatory obligations that the Abyss Navigator Suite must satisfy. These requirements are structured hierarchically in accordance with the supplied CRaM requirements tree, progressing from business context and constraints (Level 0) to required operational capabilities (Levels 1 and 2). A visual representation of this hierarchy is provided in **Appendix A**.

5.2.1 Level 0, Business Context, Constraints, Regulations, and Assumptions

BO-L0-C1 **Unmanned Operations Context**
Deep-sea operations shall be conducted without human presence at extreme depths.

BO-L0-C2 **Safety-First Business Policy**
Human exposure to extreme deep-sea environments shall be eliminated as a business policy.

BO-L0-C3 **Extreme Operating Environment Constraint**
Operations shall take place in ultra-deep-sea conditions characterised by extreme pressure, low visibility, and limited communication.

BO-L0-C4 **Regulatory Obligation**
Operations shall comply with applicable maritime, safety, and data-governance regulations.

BO-L0-A1 **Operational Assumption**
It is assumed that missions are supervised by trained and authorised personnel.

5.2.2 Level 1, Business Operational Requirements

BO-L1-1 **Autonomous Mission Execution**
The business requires missions to be executed without continuous human control.

BO-L1-2 **Remote Mission Oversight**
The business requires the ability to oversee missions from a remote operational environment.

BO-L1-3 **Fleet-Level Operations**
The business requires coordinated operation of multiple unmanned submersibles.

BO-L1-4 **Emergency and Abnormal Situation Handling**
The business requires appropriate handling of abnormal or emergency conditions during missions.

BO-L1-5 **Mission Continuity and Recovery**
The business requires missions to continue safely or terminate in an orderly manner when conditions degrade.
BO-L1-6 **Operational Accountability**
The business requires accountability for mission activities and operational decisions.

BO-L1-7 **Mission Data Value Delivery**
The business requires mission outputs to be usable for operational and analytical purposes.

5.2.3 Level 2, Business Requirement Elaborations

BO-L2-4.1
Abnormal operating conditions shall be identifiable during mission execution.

BO-L2-4.2
Missions shall transition into a safe operational state when critical conditions arise.

BO-L2-6.1
Mission activities and key decisions shall be traceable after execution.

BO-L2-6.2
Mission records shall support post-mission review and investigation.

5.3 IT / Technical Requirements Tree

The IT Technical requirements define the information-technology capabilities and constraints necessary to support the business operations of the Abyss Navigator Suite. These requirements are expressed at a high level and avoid prescribing specific technical solutions. The hierarchical structure of these requirements is illustrated in **Appendix B**.

5.3.1 Level 0, IT Context, Constraints, and Assumptions

IT-L0-C1 **Remote Operations IT Context**
Mission supervision and control shall be performed remotely from a surface or mothership environment.

IT-L0-C2 **Communications Constraint**
IT solutions shall operate under conditions of constrained bandwidth, latency, and intermittent
connectivity.

IT-L0-C3 **Security and Governance Constraint**
Operational data shall be protected in accordance with applicable security and data-governance obligations.

IT-L0-A1 **Integration Assumption**
It is assumed that primary mission sensors are supplied by the customer and made available for integration.

5.3.2 Level 1, IT Capability Requirements

IT-L1-1 **Remote Monitoring Capability**
IT capability is required to support remote monitoring of mission and vehicle status.

IT-L1-2 **Remote Intervention Capability**
IT capability is required to support remote intervention when autonomous operation is insufficient.

IT-L1-3 **Multi-Vehicle Supervision Support**
IT capability is required to support supervision of more than one unmanned submersible concurrently.

IT-L1-4 **Data Protection Capability**
IT capability is required to protect the confidentiality and integrity of operational data.

IT-L1-5 **Operational Logging Capability**
IT capability is required to support logging of mission events and operator actions.

IT-L1-6 **Degraded Communication Handling**
IT capability is required to support safe operation under degraded or intermittent communication conditions.

5.3.3 Level 2, IT Requirement Elaborations

IT-L2-5.1
Logged information shall be sufficient to support post-mission review and accountability.

6. Business Operations Requirements

This section defines our **business-level operational requirements**. These requirements describe what we require in order to operate effectively, without prescribing how solutions are implemented. Precedence indices are applied to reflect business criticality.

6.1 Autonomous Mission Execution (Essential)

We require deep-sea missions to be executed without continuous human control in order to reduce operational risk and dependence on real-time intervention.

- Missions progress through defined operational phases without continuous human intervention.
- Mission objectives are achievable without onboard human decision-making.

6.2 Remote Mission Oversight (Essential)

We require the ability to oversee unmanned missions from a remote operational environment in order to maintain situational awareness and decision authority.

6.3 Fleet-Level Operations (Desirable)

We anticipate operating multiple unmanned submersibles and therefore require coordinated oversight of concurrent missions.

6.4 Emergency and Abnormal Situation Handling (Essential)

We require predictable and safe handling of abnormal or emergency situations during missions.

- Abnormal operating conditions are identifiable during mission execution.
- Missions transition into a safe operational state when critical conditions arise.

6.5 Mission Continuity and Recovery (Essential)

We require missions to continue safely where feasible and to terminate in an orderly and recoverable manner when conditions degrade.

6.6 Operational Accountability (Essential)

We require accountability for mission activities and operational decisions.

- Mission activities and key decisions are traceable after execution.
- Mission records support post-mission review and investigation.

6.7 Mission Data Value Delivery (Desirable)

We require mission outputs to be usable for operational decision-making and analytical purposes.

6.8 External Business Systems (Acceptable)

We operate in coordination with external business systems, including regulatory reporting bodies, maritime authorities, and downstream data analysis environments. Proposed solutions are expected to support interaction with such external business systems without assuming direct control over them.

7. IT / Technical Requirements

This section defines the minimum enabling IT capabilities required to support our business operations. These requirements are intentionally high-level and non-prescriptive to avoid constraining proposed solutions.

7.1 Company Platform

We do not mandate a specific software platform. Proposed solutions must support remote operations from a surface or mothership environment and operate within the stated IT constraints.

7.2 Data Management

Proposed solutions shall support structured storage, protection, and retrieval of operational and mission data sufficient to meet accountability, review, and regulatory obligations.

7.3 Networking and Interfaces

Proposed solutions shall support communication and interfacing suitable for deep-sea operational environments characterised by constrained bandwidth, latency, and intermittent connectivity. Integration with systems supplied by us shall be supported at the software level.

7.4 IT Capability Requirements

We require IT capabilities sufficient to:

- support remote monitoring of mission and vehicle status,
- allow remote intervention when autonomous operation is insufficient,
- enable supervision of multiple unmanned submersibles concurrently,
- protect the confidentiality and integrity of operational data,
- log mission events and operator actions to support accountability,
- support safe operation under degraded or intermittent communication conditions.

8. Requirements Models

The requirements models associated with this Project Specification are provided to support contextual understanding of stakeholder structure and high-level operational interactions relevant to the Abyss Navigator Suite. These models are illustrative in nature and are not intended to prescribe system design, architecture, or implementation decisions.

An organisational context model is used to identify internal stakeholder appointments and reporting relationships relevant to the project. This model supports interpretation of stakeholder roles referenced elsewhere in this specification.

A high-level use case model is used to illustrate representative operational interactions between human operators and the Abyss Navigator Suite. The use case model is intended to provide situational context for the business requirements and does not define detailed functional behaviour.

Supporting visual representations of these models are provided in **Appendix C**.

9. Financials

We provide the following indicative financial information to communicate the expected scale of the project for tender evaluation purposes. These figures are non-contractual, are provided for comparative assessment only, and are subject to refinement following vendor selection.

9.1 Estimated Project Cost

Based on industry benchmarks for safety-critical unmanned maritime software systems of comparable scope and risk profile, the estimated project cost is USD 10-12 million.
This estimate:

- excludes physical submersible hardware and mission sensors, which are supplied separately,
- reflects software development, integration, verification, and deployment activities,
- is redacted in the released tender version to avoid biasing vendor proposals.

9.2 Indicative Payment Schedule

The following payment schedule is indicative and expressed as percentages of total contract value:

- 15% upon contract award and project mobilisation
- 25% upon delivery of core autonomous mission capability
- 25% upon completion of integration and operational readiness capability
- 20% upon successful system and fleet-level acceptance testing
- 15% upon final deployment and handover

Final payment terms will be agreed following the contract award.

10. Schedule and Milestones

We define the following indicative schedule to reflect the anticipated duration and major phases of the project. This schedule recognises the complexity and safety-critical nature of the system and is intended to guide, rather than constrain, vendor proposals.

- **Months 0–3**: Requirements confirmation, planning, and mobilisation
- **Months 4–10**: Core mission capability development and control environment establishment
- **Months 11–16**: Integration of operational subsystems and data interfaces
- **Months 17–22**: Safety, monitoring, and abnormal-condition handling capability
- **Months 23–28**: System integration, verification, and validation activities
- **Months 29–36**: Sea trials, fleet acceptance testing, regulatory readiness, deployment, and handover

Vendors may propose alternative schedules provided that overall project objectives and safety obligations are satisfied.

11. Addenda

11.1 Project Direction Change Notice

Triton Deepwater formally records this addendum to notify stakeholders and bidders of significant changes from the original Product Description. The decision to abandon manned deep-sea operations in favour of a fully unmanned, autonomous-first model was taken on the basis of safety, feasibility, and alignment with industry best practice.

12. Sign-Off

This Project Specification is certified as complete and ready for submission to vendor executive stakeholders following completion of requirements elicitation and analysis activities.

Signed by:
Sarah Whitman
Project Leader

13. Declarations

ChatGPT was used to help the team with language, grammar and flow. Our team also used it as a way to explain complex concepts to us. All content, requirements, models and diagrams were developed by the team.

Name of Team Members:

Name	**Student ID**	**Signature**
Muhammad Ridwan Putra Jasni	2401684	Ridwan

Chng Zong Han	2401892	Zong Han
Lin Yuhao	2400703	Yuhao
Muhammad Mikhail Bin Mazlan	2402298	Mikhail
Kannan s/o Rajamohan	2401517	Kannan

Appendix

Appendix A - Biz-Ops Requirements Tree

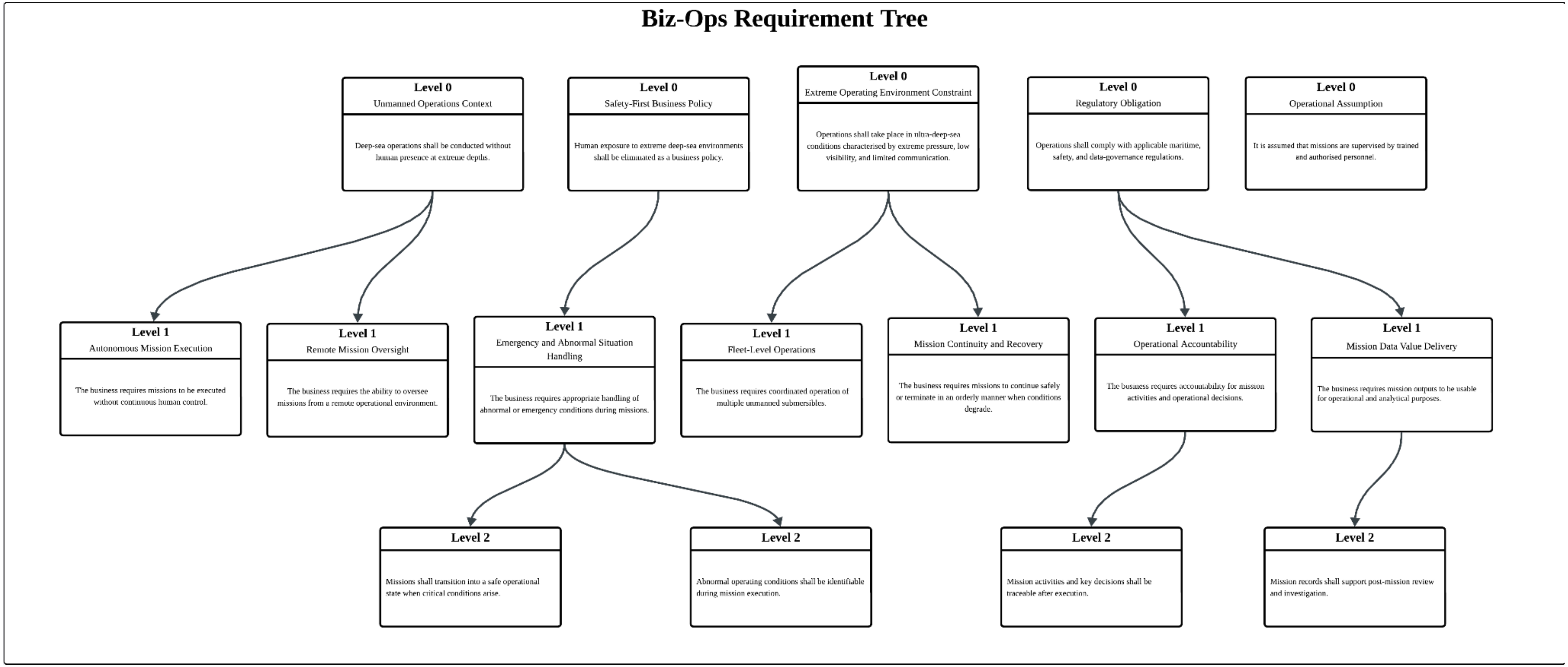

Figure A-1: Business Operation Requirements Tree for Abyss Navigator Suite Project

Appendix B - IT Tech Requirements Tree

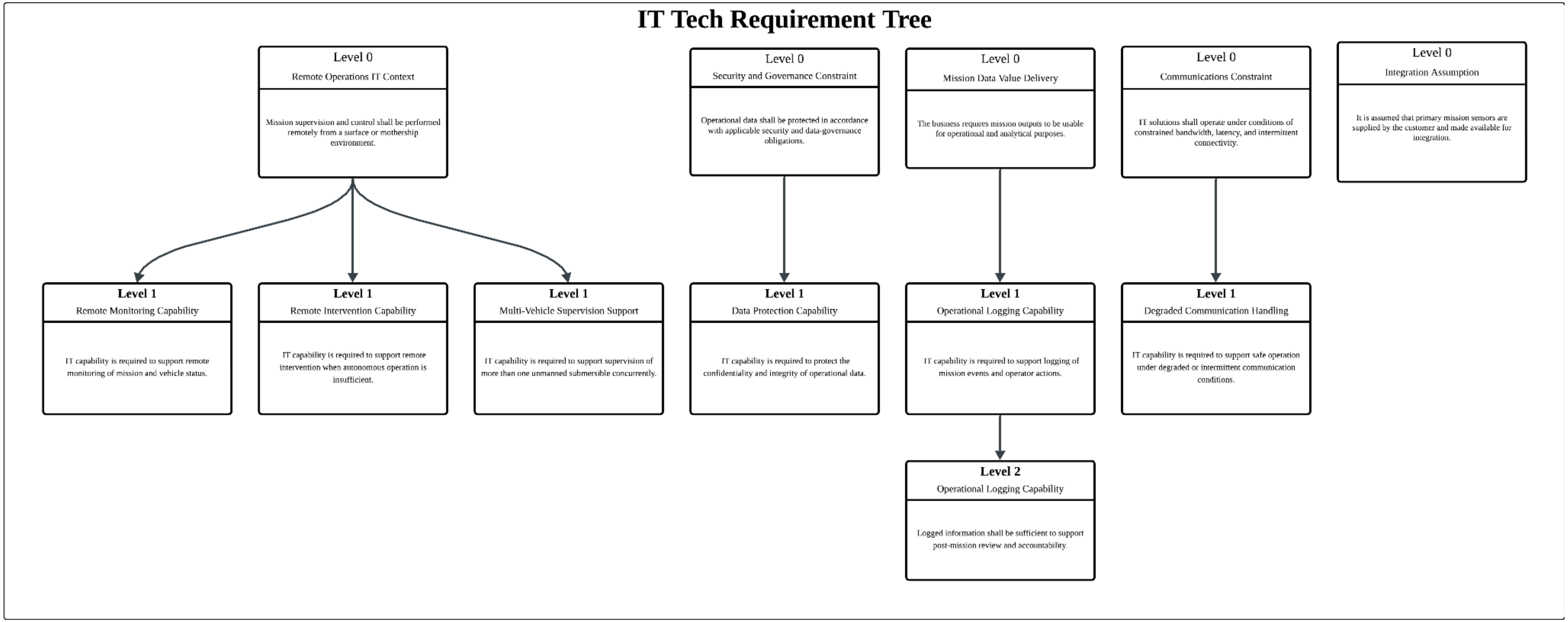

Figure B-1: IT Technical Requirements Tree for Abyss Navigator Suite Project

Appendix C - Requirements Models

Appendix C.1 - Use Case Diagram

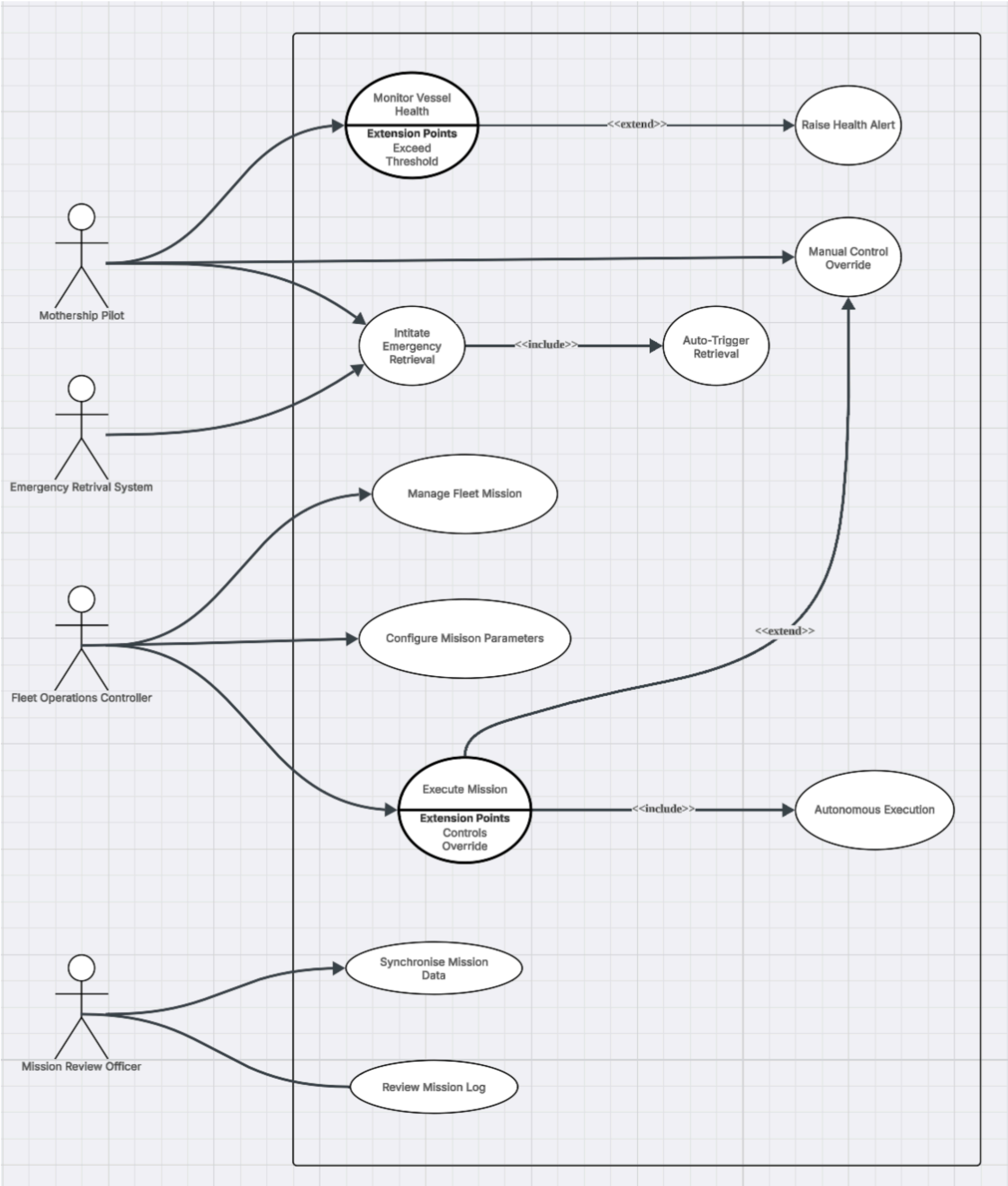

Figure C-1: Organisation Chart for Triton Deepwater Stakeholders
This organisation chart illustrates internal organisational appointments and reporting relationships relevant to the Abyss Navigator Suite project. The chart reflects permanent stakeholder roles within Triton Deepwater that influence or are impacted by the project.

Appendix C.2 - Organisation Chart

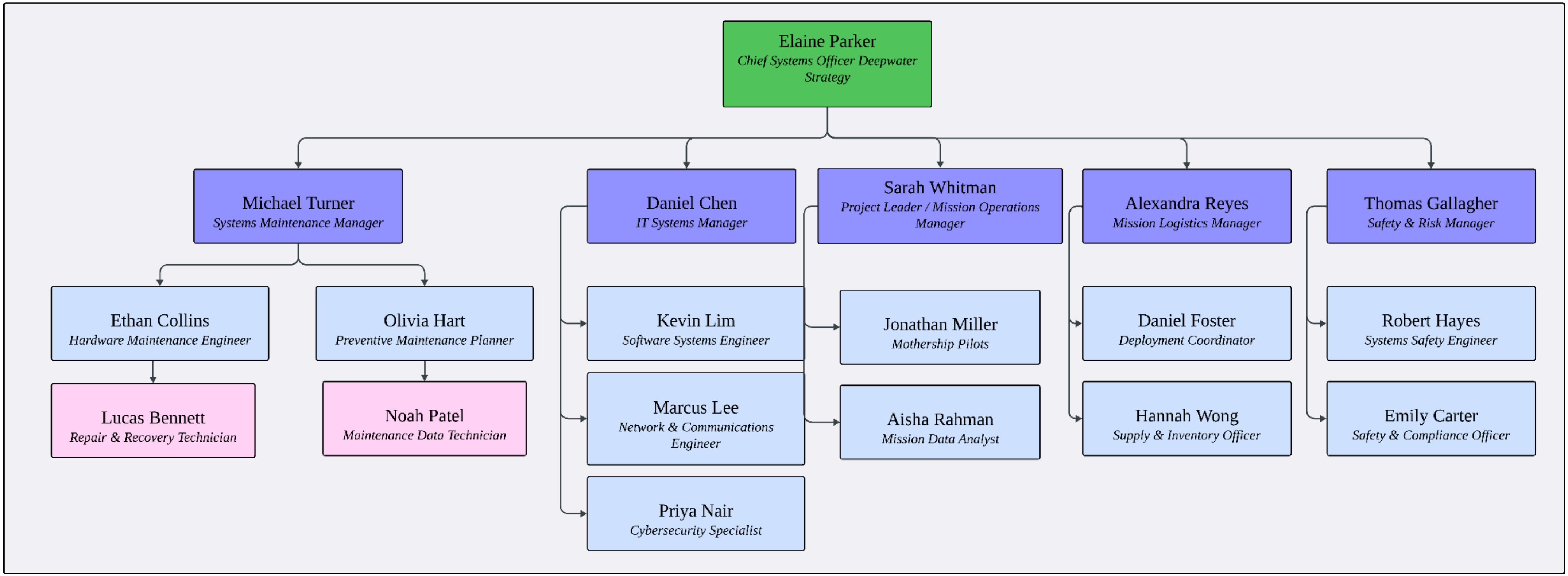

Figure C-2: Organisation Chart for Triton Deepwater Stakeholders
This organisation chart illustrates internal organisational appointments and reporting relationships relevant to the Abyss Navigator Suite project.
The chart reflects permanent stakeholder roles within Triton Deepwater that influence or are impacted by the project.

Appendix C.3 - Organisational Roles and Responsibilities

Division	Role	Name	Primary Responsibilities
Executive	Chief Systems Officer (Deepwater Strategy)	Elaine Parker	Provides overall strategic leadership and oversight of the Abyss Navigator Suite Project. Oversees all technical, operational, safety, and project activities. Approves mission readiness, system changes, and risk acceptance.
Systems Maintenance	Systems Maintenance Manager	Michael Turner	Oversees system reliability, maintenance planning, and long-term sustainment of all submersible systems under the project.
Systems Maintenance	Hardware Maintenance Engineer	Ethan Collins	Maintains and repairs physical submersible components, including propulsion, power, and structural systems.
Systems Maintenance	Repair & Recovery Technician	Lucas Bennett	Conducts repairs and recovery operations following system faults or failures.
Systems Maintenance	Preventive Maintenance Planner	Olivia Hart	Plans and schedules preventive maintenance based on system usage and health data to minimise downtime.
Systems Maintenance	Maintenance Data Technician	Noah Patel	Conducts maintenance checks according to the preventive maintenance plan and records system health data.
IT Systems	IT Systems Manager	Daniel Chen	Manages all software platforms, communication networks, and IT infrastructure supporting mission operations.
IT Systems	Software Systems Engineer	Kevin Lim	Develops and maintains mission control software and fleet management systems.
IT Systems	Network & Communications Engineer	Marcus Lee	Manages telemetry and communication links between submersibles and mothership.
IT Systems	Cybersecurity Specialist	Priya Nair	Ensures system and data security, monitors vulnerabilities, and enforces cybersecurity policies.

Project & Operations	Project Leader / Mission Operations Manager	Sarah Whitman	Responsible for overall coordination of the Abyss Navigator Suite project, including cross-team alignment, schedule oversight, and management of operational dependencies during mission planning and execution.
Project & Operations	Mothership Pilot	Jonathan Miller	Operates and navigates the mothership supporting submersible deployment and recovery.
Project & Operations	Mission Data Analyst	Aisha Rahman	Monitors mission telemetry, identifies anomalies, and supports real-time and post-mission analysis.
Mission Logistics	Mission Logistics Manager	Alexandra Reyes	Oversees logistical planning, coordination, and resource availability for missions.
Mission Logistics	Deployment Coordinator	Daniel Foster	Coordinates submersible deployment, recovery, and mission timelines.
Mission Logistics	Supply & Inventory Officer	Hannah Wong	Manages inventory, spare parts, tools, and consumables required for missions and maintenance.
Safety & Risk	Safety & Risk Manager	Thomas Gallagher	Oversees risk assessment, system safety, and regulatory compliance across all missions.
Safety & Risk	Systems Safety Engineer	Robert Hayes	Identifies system hazards and failure modes; supports safety assurance and mitigation planning.
Safety & Risk	Safety & Compliance Officer	Emily Carter	Ensures compliance with maritime safety, operational, and data protection regulations.

Table C-1: Organisational Roles Relevant to the Abyss Navigator Suite
This table summarises the responsibilities of organisational roles shown in Figure C-1 as they relate to operational oversight, safety governance, technical support, and mission execution.